Toporia library: Hashin-Shtrikman update

Integration branch | 27 October 2026 | 114 implementations: 88 selectable plugins, nine physics engines and 17 presets. Twelve models x 23 updaters plus three whole methods give 279 method names; compatibility still applies.

Elastic upper envelope at the scenario Poisson ratio

The new hashin_shtrikman law binds the scenario's actual Poisson ratio and dimension. It takes the stricter of the elastic bulk and shear HS upper limits along the solver's fixed-nu ray. For either channel M, q_M=(M0-Mmin)/(Mmin+a_M); q=max(q_bulk,q_shear); E(rho)=Emin+(E0-Emin)*rho/(1+q*(1-rho)). The finite soft phase participates in the bound, rather than being added on top of the zero-stiffness-void formula.

Plane stress uses k=E/[2(1-nu)], g=E/[2(1+nu)], a_k=g and a_g=k*g/(k+2*g). Solid 3-D elasticity uses k=E/[3(1-2*nu)], a_k=4*g/3 and a_g=g*(9*k+8*g)/(6*(k+2*g)). Shifts use the solid phase. Stiffness derivatives are analytic, and the volume budget stays geometric. Original Q4/H8 numerical sources are unchanged.

Scope and integration

This is a fixed-nu upper envelope, not the density-varying Poisson ratio in Bendsoe and Sigmund equation 17. The plane-stress nu=1/3, Emin=0 case reduces to E0*rho/(3-2*rho). Both upper-channel limits are enforced. Lower bounds, joint attainability, arbitrary phase Poisson ratios and microstructure realisability are not certified. A finite soft phase and extreme nu can put a fixed-nu tensor outside a lower-channel bound.

Optional setup and compatibility hooks leave ordinary laws unchanged. Thermal pairings are refused in preflight and at direct model initialisation: elastic bulk/shear limits are not conductivity limits. Direct Python users must call law.setup(problem). The accepted range is -1 < nu < 0.5, 0 <= Emin <= E0, with E0 positive. There is no separate penalty knob.

Verification and bounded benchmarks

Seventy-seven focused cases cover independent reciprocal-form HS bounds across six nu values, five stiffness contrasts and both dimensions, plus physical/filter adjoints, context binding, the special case and thermal preflight. The focused/interpolation/unchanged-golden group passes 106 tests. Wider schema/study and GUI checks also exercise the new law.

At nu=0.3, Emin=1e-9, density radius 1.5 and 300 MMA steps (move=0.05), MBB compliance improves from 513.19396 to 199.62258, with volume 0.499988 against 0.5. Cantilever improves from 222.04768 to 83.15603, with volume 0.399769 against 0.4. Gray fractions remain 34.83% / 26.72%; final changes are 0.002999 / 0.050000.

Both runs reach the iteration cap. These are improvement and budget checks, not convergence, binary-design or fabrication claims. The comparison study's common SIMP p=3 referee uses a different material interpolation. Full measurements and limits are recorded in experiments/README.md.

Choose Hashin-Shtrikman upper envelope (fixed nu) in Material law, or interpolation.type=hashin_shtrikman. Reusable no-output recipe: python -m experiments.material_laws --law hashin_shtrikman.

Primary sources: Bendsoe and Sigmund (1999), sections 3.2 and 3.4; Berryman (2005), 3-D HS formulas. Earlier integration updates and linked catalogue follow.

Toporia library: SINH update

Integration branch | 27 October 2026 | 113 implementations: 87 selectable plugins, nine physics engines and 17 presets.

Linear stiffness, penalised effective volume

Basic SINH now supplies two material measures: eta1=rho for stiffness and eta2=1-sinh(p*(1-rho))/sinh(p) for effective volume. The p=0 limit is linear in both. At intermediate density eta2 exceeds rho, taxing gray material through the budget rather than penalising its stiffness. The default p=3 can be continued through interpolation.penal.

The implemented variant uses the same density field in both measures: Bruns Table 2 case 5 with shared density filtering, or case 4 without filters. The resource-only filtered case 6 and hybrid stiffness penalty are not implemented. Shared filtering can retain diffuse gray boundaries; this does not reproduce the paper case-6 results.

Budget and adjoint integration

Identity volume hooks preserve the other material laws. Assembled models apply the selected effective cost in evaluations, cheap updater bisections and volume objectives. Its exact derivative is carried through filters and design representations, in 2-D/3-D. Element-density starts are calibrated to the effective budget; other representations rely on the updater. Passive solids cost one and voids cost zero.

Geometric density remains the displayed field and usual volume history/study column. effective_volume_fraction records the current design's budget cost separately. Limit checks use it and label the effective-volume budget explicitly. Compare Methods adds effective_volume while keeping its common SIMP p=3 referee; the referee and SINH linear-material objective measure different material models. Effective cost is not geometric mass or a homogenisation certificate.

Bounded validation and measured limits

Twenty-four focused cases cover the material equations, 2-D/3-D physical/filter/network adjoints, five optimiser paths, continuation, variants, pinned material, volume objectives and limit reporting. The focused/interpolation/unchanged-golden group passes 53 tests. Comparison-study and GUI reachability checks also exercise the new law.

With 300 MMA steps, p=3, move=0.05, density radius 1.5 and 60-cell-wide grids, MBB compliance improves from 550.63805 to 202.79554. Its geometric fraction is 0.414981 and effective cost 0.500000 against limit 0.5. Cantilever improves from 234.23086 to 84.58895, with geometric fraction 0.337000 and effective cost 0.398823 against 0.4.

Both runs stop at the cap. Gray fractions are 33.75% / 27.22%, with final changes 0.007759 / 0.038767. These are bounded improvement and effective-budget checks, not convergence, binary designs or direct performance comparisons to penalised SIMP. Fresh final responses and full scope are recorded in experiments/README.md.

Use Material law: SINH (linear stiffness, penalised volume), or interpolation.type=sinh. Reusable no-output recipe: python -m experiments.material_laws --iterations 300 --mesh 1.

Primary source: Bruns (2005), equations 22-23 and Table 2; readable paper. Earlier integration updates and linked catalogue follow.

Toporia library: neural field update

Integration branch | 26 October 2026 | 112 implementations: 86 selectable plugins, nine physics engines and 17 presets.

A coordinate network as the design representation

The new neural representation maps normalised physical coordinates to density through dense tanh hidden layers and a sigmoid output. Every weight and bias is an optimisation variable. Explicit NumPy reverse propagation carries the exact physical/filter sensitivities back to these variables. It supports 2-D/3-D and needs no neural runtime, dataset or pretrained weights.

The default network has two hidden layers of width 20: 501 parameters in 2-D, independent of mesh size. Local seeded initialisation preserves reproducibility without altering the global random generator. Output weights start at zero; a logit bias gives the requested uniform starting volume. Finite symmetric weight/bias bounds and initial-volume checks are explicit.

Independent TOuNN-style variant

This is an independent smooth variant, not the original PyTorch implementation. Batch-normalised LeakyReLU, the density offset, AMSGrad, Fourier features and built-in symmetry/manufacturing rules are absent. No GPL source code was copied. Existing Adam uses an inequality augmented Lagrangian; general-vector MMA also works. Element-specific OC/BESO/SiMPL updaters are refused before running.

Measured behaviour and limits

On 600-step, seed-0 MBB / cantilever runs, Adam improves compliance from 984.55480 / 609.61732 to 389.76771 / 192.14204, at volumes 0.499926 / 0.399894 (limits 0.5 / 0.4). MMA reaches 346.15640 / 144.21322 at volumes 0.499657 / 0.399825. Both use the same SIMP p=3 and density radius 1.5. Adam lr is 0.01 with decay 0.995; MMA move is 0.005.

These runs stop at the iteration cap. Gray fractions remain 83.33% / 64.89% for Adam and 82.33% / 38.00% for MMA. This is functional integration and bounded improvement, not convergence, crispness, certified length scale or reproduction of reference TOuNN performance. Small weight movement alone cannot establish an adequate design.

Sixteen focused tests cover all weight/bias derivatives in 2-D/3-D, the filtered FE chain, deterministic starts, bounds, mesh-independent parameter count and updater preflight. Forty combined representation tests and 53 selected GUI form/reachability cases pass. Existing schema/composition/golden regressions are also checked.

Use the GUI Design representation panel, or representation.type=neural with q4+adam (lr=0.01) or q4+mma (move=0.005). Reusable recipe: experiments/neural.py; full results and scope are in the representation and experiments READMEs.

Source: Chandrasekhar and Suresh (2021), TOuNN. Reference repository. Earlier integration updates and the linked catalogue follow.

Toporia library: multiresolution update

Integration branch | 26 October 2026

111 implementations: 85 selectable plugins, nine physics engines and 17 presets. Twelve models and 23 updaters form 276 compositions; three whole methods bring the method-name total to 279.

Nested Q4 analysis with fine density

The new q4_multiresolution model keeps the design grid fine and solves a nested coarse bilinear displacement space. Stiffness uses exact 2 x 2 Gauss integration in each constant-density fine cell. Fine loads transfer by virtual work; fine supports become interpolation constraints. Complete compliance gradients include all fine-cell contributions. Factor 1 retains the unchanged Q4 reference path.

Scope: aligned, divisible 2-D Cartesian meshes and elastic compliance. This is an independent MTOP variant, not the original centroid-rule implementation, an adaptive/high-order method or an independent three-mesh system. Stress, springs and 3-D are outside this model.

Bounded benchmark: 100 OC steps

Preset / factor	System DOFs	Coarse C	Fresh fine C	Volume / limit
MBB / 1	2540	249.35203	249.35203	0.499996 / 0.5
MBB / 2	670	244.09112	249.59705	0.500019 / 0.5
Cantilever / 1	3720	110.01908	110.01908	0.399984 / 0.4
Cantilever / 2	960	108.08059	110.14153	0.399985 / 0.4

Factor 2 underestimates fine compliance by 2.21% / 1.87%. Single-run optimisation timings were 2.125 / 1.099 s (MBB factors 1 / 2) and 3.884 / 1.675 s (cantilever); these illustrate one machine/run, not general speed guarantees. Runs are unconverged. MBB factor 2 has a small OC volume overshoot.

Verification and known limits

The broad model/composition/schema/golden group passed 987 tests and selected GUI checks passed 102. Eighteen focused cases include independent subcell quadrature, rectangular cells, multiple loads, off-coarse supports and finite differences. A disconnected fine-grid cut deliberately exposes artificially stiff coarse designs. Filtering does not certify removal of QR-patterns. Compare Methods keeps its independent fine-Q4 referee.

Reusable recipe: experiments/multiresolution.py. Implementation and limits: toporia/plugins/models/README.md and physics/README.md.

Sources: Nguyen et al. (2010), MTOP; Gupta et al. (2018), QR-pattern artefacts. Earlier update pages and the linked catalogue follow.

Projection-based self-support

Gaynor–Guest / independent 2-D cell discretisation

Current catalogue: 84 selectable plugins + 8 physics engines + 17 presets = 109 implementations. The method-name total remains 256. This update supersedes earlier totals and remaining-work lists in the following snapshots.

Method and use

The **self_support** filter multiplies the design field by a projected average of the supported field inside a lower wedge, then applies a local cone average and exponential projection. An exact reverse sweep carries sensitivities through every lower-layer dependency. Both projection sharpnesses support continuation.

Select **Gaynor–Guest self-support projection (2-D)** as the last filter. It includes local regularisation. Use element densities, a rectangular 2-D box and no forced solid/void regions. The first cell layer touches the build plate; +y builds upwards. The angle is measured from the plate, so 90 degrees permits vertical supports only.

Scope: cell-centred points replace the paper's nodal examples. The support threshold depends on the discrete grid and angle. Finite smoothing and boundary truncation do not certify an exact angle, minimum feature size or printing process.

Validation

434 manufacturing/schema/variant checks, 53 GUI checks and 16 unchanged golden cases pass. The preceding full suite after machining passed 2363 tests with one optional multigrid skip. Small move limits down to 0.00001 now survive GUI/schema round trips; the existing default stays 0.2.

The following 500-iteration GCMMA runs use RAMP q 10 to 30 and local sharpness 10 to 25, on better-resolved meshes. Initial and final compliance use the same final material law. Neither run is claimed to have converged.

Benchmark	Initial to final compliance	Physical volume / limit
MBB / 60 x 20	12522.25648 to 454.66344	0.499995 / 0.5
Cantilever / 60 x 30	2901.78844 to 296.54728	0.399999 / 0.4

Final changes remain 0.02508 / 0.01052 and gray fractions 20.50% / 15.83%. Thresholding at 0.5 raises volumes to 0.519167 / 0.411111, above budget. An independent one-layer 45-degree check finds 0 / 1 unsupported cells. Binary designs are not labelled feasible or printable; the report also records their fresh compliance.

Reproduce and inspect

Run **python -m experiments.self_support --iterations 500 --mesh 1 --updater pymoto_gcmma** (optional pyMOTO). The default script also reproduces the own-MMA small-move trial. Progress and measurements go to the console; no saved run outputs are generated. The README records oscillation, grayness and thresholding limits.

Gaynor & Guest, SMO 54 (2016) 1157–1172 / implementation. New source files need the local integration checkout until pushed. No published code is copied.

Next: multiresolution meshes, TOuNN, SINH / Hashin–Shtrikman laws, and original-code references where a runtime is available. Machining is committed as 4decae8. Each addition remains separate; GUI and launcher source checkouts are unchanged.

Machining accessibility

26 October 2026 / integration/physics

Current catalogue: 83 selectable plugins + 8 physics engines + 17 presets = 108 implementations. Eleven models and 23 updaters form 253 compositions; with three whole methods, there are 256 method names.

New method: PDE machining

The **machining** filter is an independent Cartesian variant of the Høghøj–Träff advection–diffusion formulation. It intersects material shadows for selected tool approach sides in 2-D or 3-D and provides the complete gradient through the PDE, intersection and projection. Passive solids cast shadows.

Use **Element densities**, a density/PDE regulariser, then **PDE machining (axis directions)** as the last filter. A tool from +y approaches from above. Forced-void regions are refused. Initial source densities meet the physical volume budget; convergence also monitors physical-density changes.

Recipe: MMA; move 0.01; initial asymptotes 0.03; growth 1.05; shrink 0.65; tolerance 0.00001. Defaults: Peclet 10000, projection beta 8, threshold 0.4 and shadow power -4. Large optimiser moves can be unstable.

Validation and scope

Twenty-five focused numerical tests pass. The 80-iteration coarse-grid runs below reevaluate the final design; neither is claimed to have converged. Independent 0.5-threshold ray checks found zero inaccessible void cells for the selected tool side.

Benchmark	Compliance: initial to final	Physical volume
MBB / 24 x 8	217476025.72 to 686.42380	0.499864 / 0.5
3-D cantilever / 24 x 8 x 4	648.33247 to 76.06831	0.299961 / 0.3

Limits: Cartesian directions and ideal zero-width tools. No finite cutter, holder collision or minimum-size guarantee. Diffusion, projection and gray densities can leave accessibility artefacts. This is not a port of Langelaar's finite-tool method or a machining certificate.

Other integration fixes

Small mechanism loads retain their precision. Compare Methods selects compatible physics and objectives; all 17 presets have GUI-to-study regressions, plus a 321-combination compatible-updater smoke audit. These short audits check execution, not convergence.

Source and continuation

Høghøj & Träff, CMAME 391 (2022) 114488 / readable full paper. Local code: machining.py (new files need the local checkout until pushed). Reproduce: python -m experiments.machining --iterations 80. The script prints measurements without saving runs. Full regression: 2363 tests passed; one optional multigrid check skipped.

Next: Gaynor–Guest self-support; multiresolution meshes; TOuNN dependency check and integration; SINH / Hashin–Shtrikman laws; original-code references where a runtime is available. Each addition is completed and committed separately.

This update takes precedence over the earlier 13-page catalogue snapshot that follows (checkpoint 9abc824). Its completed items and historical sources are preserved. GUI and launcher source checkouts remain separate. No code is pushed or merged.

The library

What is implemented, what changed, and what remains

A linked catalogue of the current integration checkout, updated with the work completed through minimum-length constraints.

82	8	17
selectable plugins	physics engines	benchmark presets

107 implementations in total. The 11 models and 23 updaters form 253 named compositions, plus three whole methods: 256 method names.

26 October 2026
Branch `integration/physics` / implementation checkpoint `9abc824`

Completed work / Current catalogue / Remaining work

Work is isolated in the integration checkout. The GUI and launcher source checkouts are unchanged; the owner reviews and merges later.

Paper and repository links are clickable. Source-code links target integration/physics; newly committed files may require the local checkout until the branch is pushed. This document does not publish or merge code.

Completed integration work

The small and medium lists from the original PDF define this round. Existing physics was stabilised and checked before new methods were added. Each row below has its own commit; restricted variants are identified explicitly.

Addition	Implemented scope / result	Commit
Existing physics stabilised	Repaired compatible compositions, scaling, constraint guards and numerical issues. Existing thermal, dynamic and nonlinear physics remain in the catalogue.	2dc42a1
Q4 linearised buckling	Recovered and validated the eigenproblem, complete adjoint and column benchmark; sparse solution and deterministic mode handling.	b9f7a7f
Anisotropic Helmholtz filter	Conservative 2-D/3-D finite-volume filter with directional radii, preserved constants/volume and an exact adjoint.	ce9f4e2
Moving morphable voids	Restricted 2-D superelliptical voids through the MMC kernel. No 3-D B-spline surfaces or FE pruning.	6a10f2b
Floating projection	Updater with mass-preserving threshold, fixed-region/bound handling and internal beta continuation. The original backlog's 'without beta continuation' description is corrected.	00ffa9b
Annealing evaluation	Own bounded standard-SA experiment; not an adaptive topASA port. It missed the unchanged 10%-of-OC gate on three seeds at 200 and 2000 solves and is not registered.	8116acf

These are independently implemented paper variants unless the catalogue explicitly identifies a port or an external-package wrapper. No original code was copied where reuse permission was unclear.

Completed work / continued

Addition	Implemented scope / result	Commit
Compliant mechanisms	Physical input/output ground springs, signed output-displacement adjoint, a dedicated Q4 model and Force Inverter preset; pyMOTO cross-check.	4496e90
Geometry projection	Independent 2-D/3-D fixed-radius capsule bars with endpoint and activation variables, bounded union and complete derivatives; no GPTO code copied.	55c019e
Reaction-diffusion level set	Sparse cell-centred 2-D compliance method using the paper's elastic topological-sensitivity tensor, an implicit PDE step and updated-volume control.	c76509f
Hamilton-Jacobi level set	Godunov evolution with circular-hole plane-stress sensitivity, hole nucleation, reinitialisation and volume feedback. Not a standalone Amstutz geodesic solver.	a27b13b
Phase-field energy	2-D/3-D regularised-compliance objective with a physical diffuse-interface energy and exact derivatives. Uses existing optimisers; no dedicated Allen-Cahn/Cahn-Hilliard solver.	71abd18
Geometric minimum length	Solid and void constraints through a separate cone filter, complete adjoint, conservative nodal KS and fixed 1-D tolerance calibration. Not an exact geometric size certificate.	9abc824

Validation checkpoint

The latest full native run took 28 min 13 sec: **2303 passed, one optional multigrid skip and one outdated capability assertion failed**. Only that expected capability tuple was corrected; all six tests in its file passed on rerun. All **2304 current tests have passing coverage** across those runs. The original full-run log is retained, rather than described as a clean monolithic run.

The new methods have gradient, physical/geometric fixture and composed-run checks. GUI reachability, original golden checks, Ruff and whitespace checks pass. Original Q4/H8 sources and golden baselines are unchanged. MATLAB/Octave reference runs remain optional and have not been claimed as completed.

Minimum-length limitation: uniform gray starts stalled or retained violations in tested strict two-phase runs. A resolved feasible-seed cantilever with linear ersatz improved compliance from 115.115877 to 111.907874 at volume 0.5; fresh solid/void residuals were -2.22e-7 / -3.23e-8. This is a modest local validation, not evidence of reliable topology creation from a uniform start.

Detailed integration record / Reproducible validation recipes

The current catalogue

The following tables are generated from `toporia/plugins/LIBRARY.md`. They include the new entries and the already implemented physics. Source and paper references retain their clickable links. Optional packages remain separate dependencies.

Port: translated from named code. **Paper**: independently written from a paper. **Wraps**: calls an optional external package. **Own**: Toporia's own implementation or standard practice.

Design representations: what the design variables are (folder)

In the app	Name	File	Came in	From
Element densities	`element_density`	element_density.py	Own	Bendsøe 1989; Sigmund's 99-line code · papers 1 2
Moving morphable components (constant or variable thickness)	`mmc`	mmc.py	Paper	Guo, Zhang & Zhong 2014; variable thickness as in MMC188 (Zhang et al., *SMO* 2016) · papers 1 2
Moving morphable voids (restricted 2D superellipses)	`mmv`	mmv.py	Paper	Zhang et al., *CMAME* 2017; own geometric dual of MMC, without B-spline surfaces or FE pruning · paper
Geometry projection (2D/3D capsule bars)	`geometry_projection`	geometry_projection.py	Paper	Own fixed-radius, bounded-union variant with component activation; no GPTO code copied · papers 1 2
B-spline field	`bspline`	bspline.py	Paper	Qian, *CMAME* 2013 · paper

Filters: design to physical density, and fabrication rules (folder)

In the app	Name	File	Came in	From
Density filter	`density`	density.py	Port	top88 (Andreassen et al. 2011); Bruns & Tortorelli 2001; Bourdin 2001 · papers 1 2 3
PDE (Helmholtz) filter	`pde`	pde.py	Port	the PDE variant of top88 (top82.m); Lazarov & Sigmund 2011 · papers 1 2
Anisotropic PDE (Helmholtz) filter	`pde_anisotropic`	pde_anisotropic.py	Paper	Directional extension of Lazarov & Sigmund 2011, using conservative cell-centred finite volumes · paper
Sensitivity filter	`sensitivity`	sensitivity.py	Port	Sigmund 1997, as in the 99-line code and top88 · papers 1 2 3
Heaviside projection (fixed or volume-preserving threshold)	`heaviside`	heaviside.py	Paper	Guest et al. 2004; Wang, Lazarov & Sigmund 2011; volume-preserving as in top99neo (Ferrari & Sigmund 2020) · papers 1 2 3
Morphology filter (erode / dilate / open / close)	`morphology`	morphology.py	Paper	Sigmund, *SMO* 2007 · paper
AM overhang filter	`am`	am.py	Port	Langelaar's AMfilter.m (*SMO* 2017) · papers 1 2
Routing radius filter	`routing`	routing.py	Own	inspired by Sigmund 2007 · paper
Symmetry filter	`symmetry`	symmetry.py	Own	standard practice · paper
Extrusion (constant cross-section)	`extrusion`	extrusion.py	Paper	Ishii & Aomura 2004 · paper
Pattern repetition	`pattern`	pattern.py	Paper	Bendsøe & Sigmund 2003; Almeida, Paulino & Silva 2009 · papers 1 2
Casting (no undercuts)	`casting`	casting.py	Paper	Gersborg & Andreasen, *SMO* 2011 · paper

Material laws: density to stiffness (folder)

In the app	Name	File	Came in	From
SIMP (power law)	`simp`	simp.py	Paper	Bendsøe 1989 · paper
RAMP (rational)	`ramp`	ramp.py	Paper	Stolpe & Svanberg 2001 · paper
Ordered SIMP (multi-material)	`ordered_simp`	ordered_simp.py	Paper	Zuo & Saitou, *SMO* 2017 · paper

Physics models: what computes displacements and gradients (folder, elements in physics/)

In the app	Name	File	Came in	From
2-D Q4 plane stress (Toporia)	`q4`	q4.py, q4_plane_stress.py	Port	the physics half of top88 · paper
2-D Q4 compliant mechanism (Toporia)	`q4_mechanism`	q4_mechanism.py, q4_mechanism.py	Paper	Own port-spring Q4 engine; optional independent pyMOTO assembly/backpropagation check · paper
2-D Q4 with buckling (Toporia)	`q4_buckling`	q4_buckling.py, q4_buckling.py	Paper	Ferrari, Sigmund & Guest 2021; Toporia's p-norm aggregate and column geometry · paper
2-D Q4, large deformation (Toporia)	`q4_nonlinear`	q4_nonlinear.py, q4_nonlinear.py	Paper	Buhl, Pedersen & Sigmund, *SMO* 2000; energy interpolation after Wang, Lazarov, Sigmund & Jensen, *CMAME* 2014; open reference: Han et al. 2021 · papers 1 2 3
3-D H8 bricks (Toporia)	`h8`	h8.py, h8_solid.py	Port	top3d (Liu & Tovar, *SMO* 2014) · paper
Linear elasticity, pyMOTO network	`pymoto_elastic`	pymoto_elastic.py	Wraps	pyMOTO (MIT), `pip install pymoto`
2-D Q4 heat conduction (Toporia)	`q4_heat`	q4_heat.py, q4_heat.py	Paper	Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al., *SMO* 2006 · paper
3-D H8 heat conduction (Toporia)	`h8_heat`	h8_heat.py, h8_heat.py	Paper	the 3-D brick of the same heat problem · paper
Heat conduction, pyMOTO network	`pymoto_heat`	pymoto_heat.py	Wraps	pyMOTO's `AssemblePoisson` (MIT), `pip install pymoto`; the cross-check of `q4_heat`
2-D Q4 with vibration (Toporia)	`q4_dynamic`	q4_dynamic.py, q4_dynamic.py	Paper	Pedersen, *SMO* 2000; Du & Olhoff, *SMO* 2007 · papers 1 2
Vibration, pyMOTO network	`pymoto_dynamic`	pymoto_dynamic.py	Wraps	pyMOTO `AssembleMass` and `EigenSolve`, `pip install pymoto`

Objectives and constraints (folder)

In the app	Name	File	Came in	From
Compliance	`compliance`	compliance.py	Own	the classic objective (Bendsøe & Sigmund 2003) · paper
Phase-field regularised compliance	`phase_field`	phase_field.py	Paper	Own 2-D/3-D fixed-load diffuse-interface energy, with existing gradient optimisers; no separate PDE solver · papers 1 2
Output displacement (maximise)	`output_displacement`	output_displacement.py	Paper	Signed point output, one shared spring-augmented adjoint · paper
Material volume	`volume`	volume.py	Own	as in Le et al. 2010 and pyMOTO's stress example · paper
Geometric minimum feature length	`minimum_length`	minimum_length.py	Paper	Own 2-D/3-D physical-field variant, separate cone filter, full indicator adjoint, conservative nodal KS and fixed tolerance · papers 1 2
Local von Mises stress (augmented Lagrangian)	`local_stress`	local_stress.py	Paper	Giraldo-Londoño & Paulino, *SMO* 2021 (PolyStress); Senhora et al., *SMO* 2020 · papers 1 2
End compliance (non-linear)	`end_compliance`	end_compliance.py	Paper	Buhl, Pedersen & Sigmund, *SMO* 2000 · paper
Peak von Mises stress	`stress`	stress.py	Paper	Le, Norato, Bruns, Ha & Tortorelli, *SMO* 2010 · paper
Buckling load factor	`buckling_load`	buckling.py	Paper	Reciprocal-factor aggregate; objective or constraint on `q4_buckling` · paper
Thermal compliance	`thermal_compliance`	thermal_compliance.py	Paper	Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al., *SMO* 2006 · paper
Fundamental frequency (maximise)	`eigenfrequency`	eigenfrequency.py	Paper	Díaz & Kikuchi, *IJNME* 1992; Du & Olhoff, *SMO* 2007 · papers 1 2
Fundamental frequency (at least)	`eigenfrequency_limit`	eigenfrequency.py	Paper	the same, as a constraint
Dynamic compliance (harmonic load)	`dynamic_compliance`	dynamic_compliance.py	Paper	Ma, Kikuchi & Hagiwara, *Comput. Mech.* 1993 (no link); PolyDyna · paper

Optimisers: how the design moves (folder)

In the app	Name	File	Came in	From
Optimality criteria	`oc`	oc.py	Port	the 99-line code and top88 · papers 1 2
Floating projection (FPTO)	`floating_projection`	floating_projection.py	Paper	Huang, *Engineering Structures* 2020 and *Adv. Eng. Softw.* 2021; modular updater with internal β continuation · papers 1 2
Optimality criteria, several constraints	`oc_multi`	oc_multi.py	Own	after Zhou & Rozvany 1991 · paper
Method of moving asymptotes	`mma`	mma.py	Port	Svanberg 1987, one-constraint form of top88's MMA variant · papers 1 2
CONLIN (convex linearisation)	`conlin`	conlin.py	Paper	Fleury & Braibant 1986; Fleury 1989 · papers 1 2
Mirror descent (SiMPL)	`simpl`	simpl.py	Paper	Keith, Kim, Lazarov & Surowiec, *SMO* 2025 · paper
BESO (soft kill)	`beso`	beso.py	Paper	Huang & Xie 2007, 2010 · papers 1 2
BESO (hard kill)	`beso_hard`	beso_hard.py	Paper	Huang & Xie 2010 · papers 1 2
TOBS (integer LP)	`tobs`	tobs.py	Paper	Sivapuram & Picelli 2018; the 101-line code of Picelli et al. 2021; solved with HiGHS (SciPy) · papers 1 2
Augmented Lagrangian (local constraints)	`augmented_lagrangian`	augmented_lagrangian.py	Paper	PolyStress outer loop (Giraldo-Londoño & Paulino 2021) with MMA inner steps · papers 1 2
Adam (augmented Lagrangian)	`adam`	adam.py	Paper	Kingma & Ba 2015; the volume penalty of TOuNN (Chandrasekhar & Suresh 2021) · papers 1 2
Proportional topology optimisation	`pto`	pto.py	Paper	Biyikli & To, *PLoS ONE* 2015 · paper
MMA (pyMOTO), GCMMA (pyMOTO)	`pymoto_mma`, `pymoto_gcmma`	pymoto_optimizers.py	Wraps	pyMOTO (MIT) · papers 1 2
MMA (mmapy), GCMMA (mmapy)	`mmapy_mma`, `mmapy_gcmma`	mmapy_optimizers.py	Wraps	mmapy, Svanberg's reference code in Python (**GPL-3**, optional), `pip install mmapy` · papers 1 2
SLSQP (SciPy)	`scipy_slsqp`	scipy_slsqp.py	Wraps	SciPy (BSD), Kraft 1988
trust-constr (SciPy)	`scipy_trust_constr`	scipy_minimize.py	Wraps	SciPy (BSD) · paper
MMA, CCSAQ, SLSQP, augmented Lagrangian + L-BFGS (NLopt)	`nlopt_mma`, `nlopt_ccsaq`, `nlopt_slsqp`, `nlopt_auglag`	nlopt_optimizers.py	Wraps	NLopt (LGPL/MIT), `pip install nlopt` · papers 1 2 3 4
Null-space optimiser	`nullspace`	nullspace.py	Wraps	null-space optimizer of Feppon, Allaire & Dapogny 2020 (**GPL-3**, optional), `pip install nullspace_optimizer` · paper

Whole methods: their own loop (folder)

In the app	Name	File	Came in	From
RBF level set	`levelset`	levelset_rbf.py	Port	TOPRBF.m (Luo et al.); Wang & Wang, *CMAME* 2006 · papers 1 2
Reaction-diffusion level set	`levelset_reaction_diffusion`	levelset_reaction_diffusion.py	Paper	Otomori et al. 2015; own cell-centred variant, implicit diffusion and bisected volume multiplier · paper
Hamilton-Jacobi level set with circular-hole derivative	`levelset_hamilton_jacobi`	levelset_hamilton_jacobi.py	Paper	Challis 2010; independent upwind HJ/TD, reinitialisation and normalised augmented-Lagrangian volume feedback · paper

Every model can also be paired with every optimiser as a method called `<model>+<optimiser>`, for example `q4+nlopt_mma`. Those pairings are built on demand, so they are not listed here.

Continuation: parameters that change during a run (folder)

In the app	Name	File	Came in	From
Equal steps	`steps`	steps.py	Own	the classic SIMP continuation (Sigmund & Petersson 1998) · paper
Multiply by a factor	`geometric`	geometric.py	Own	the usual β continuation (Guest et al. 2004; Wang et al. 2011) · papers 1 2
Linear ramp	`linear`	linear.py	Own	-
When the design settles	`settled`	settled.py	Own	common practice in the projection literature (Wang et al. 2011) · paper

The robust formulation (one design evaluated in several versions) is a solver setting rather than a schedule. See the schedules README and the preset *MBB Beam (Robust)*.

Post-processing: what is made of the finished design (folder)

In the app	Name	File	Came in	From
Threshold to black and white	`threshold`	threshold.py	Own	as method comparisons do (Rojas-Labanda & Stolpe 2015) · paper
Connectivity check	`connectivity`	connectivity.py	Own	-
Feature size	`feature_size`	feature_size.py	Own	measures what Sigmund 2007 and Wang et al. 2011 promise · papers 1 2
Grey level (M_nd) over the run	`grey_level`	grey_level.py	Paper	Sigmund 2007 · paper
Export outline (SVG), (DXF)	`export_svg`, `export_dxf`	export_svg.py, export_dxf.py	Own	-
Smooth outline (SVG + DXF)	`smooth_outline`	smooth_outline.py	Paper	Subedi et al. 2020 (review); smoothing splines or Chaikin 1974 · papers 1 2
Export solid (STL)	`export_stl`	export_stl.py	Own	-

Benchmark problems (folder)

In the app	File	From
MBB Beam	mbb_beam.py	the 99-line code and top88 · papers 1 2
MBB Beam (Robust)	mbb_beam_robust.py	Wang, Lazarov & Sigmund 2011 · paper
Vibrating Beam	vibrating_beam.py	after Du & Olhoff 2007 · paper
Column Buckling	column_buckling.py	Toporia column geometry, Ferrari, Sigmund & Guest's linearised formulation · paper
Cantilever	cantilever.py	the 99-line code · paper
Nonlinear Cantilever	nonlinear_cantilever.py	Buhl, Pedersen & Sigmund 2000; Wang et al. 2014 · papers 1 2
Cantilever 3D	cantilever_3d.py	top3d (Liu & Tovar 2014) · paper
Michell Cantilever	michell_cantilever.py	Michell 1904 · paper
L-bracket (local stress)	l_bracket_stress.py	Giraldo-Londoño & Paulino 2021 (PolyStress); geometry of the L-bracket, limit chosen here · paper
L-bracket	l_bracket.py	Duysinx & Bendsøe 1998; Le et al. 2010 (dimensions still to be checked against the papers) · papers 1 2
Three-Point Bending	three_point_bending.py	the materials test
Heat Sink	heat_sink.py	Bendsøe & Sigmund 2003, section 1.5; Gersborg-Hansen et al. 2006 · paper
Standard Bar	standard_bar.py	Toporia's own
Drone Arm, Drone Arm (Camera View), Drone Arm (Point Loads)	drone_arm.py, drone_arm_camera_view.py, drone_arm_point_loads.py	Toporia's own application case
Force Inverter	force_inverter.py	Own 40 × 40 geometry and port values following Sigmund 1997; signed output and input/output ground springs · paper

Evaluated but not registered

The bounded simulated-annealing experiment misses the unchanged 10%-of-OC performance gate on all three seeds at 200 and 2000 solves. It remains reproducible outside plugin discovery; no failing part was registered and no threshold was relaxed. This is an own standard-SA baseline, not a port or evaluation of adaptive topASA. Catalogue implementation counts exclude unregistered experiments.

Remaining work

Current linear sequence: machining accessibility, projection-based self-support, multiresolution meshes, then neural reparameterisation (subject to dependency support). Machining has no code changes yet: a readable full-text copy of Langelaar (2019) is needed after institutional-download failures.

The original scope PDF remains unchanged. MATLAB/Octave oracles are optional; SINH/Hashin-Shtrikman and large framework work remain outside this agreed round.

Listed from least to most effort. "Code" means open code exists (port it, or check against it). "Paper" means it would be written from the paper. Details and status are in TODO.md.

Small: one file in an existing slot

What	Kind	Source	Notes
Reference results from the original MATLAB codes (top88, top99neo, top3d, TOBS, MMC, the level-set codes)	tests	Code: run once in GNU Octave or MATLAB · papers 1 2 3 4 5 6 7	proves the Python versions give the same answers

Medium: new physics or a whole method

What	Source
Machining accessibility, projection-based self-support	Paper: Langelaar 2019; Gaynor & Guest 2016 · papers 1 2
Multi-resolution design and analysis meshes	Paper: Nguyen et al. 2010 · paper
Neural reparameterisation (TOuNN)	Code: TOuNN (needs PyTorch or JAX, which may not install on Python 3.14) · paper
SINH and Hashin-Shtrikman material laws	Paper: Bruns 2005; Bendsøe & Sigmund 1999 (the material law has to see the volume and Poisson's ratio) · papers 1 2

Beyond this round

The original long-term research map is retained below. These larger framework changes and runtime blockers were not implemented in this integration round.

Large: the framework has to change first

What	Unlocks
Several fields per element	DMO, fibre orientation, true multi-material
Unstructured meshes	PolyTop/PolyMesher, VEM, FEniTop, CAD import
Beam and truss elements	ground structures (GRAND), Michell layouts
Multiscale, IGA, CutFEM	de-homogenisation (deHomTop808), IgaTop, GridapTopOpt
Fluids, transient, acoustics, photonics	topflow, TOFLUX, ceviche, Meep
Learned methods	TopoDiff, TopologyGAN, dl4to and the SELTO datasets

Historical runtime blockers

These runtime statements are retained from the original 24 October survey; package availability was not rechecked for this PDF update.

What	Why
IPOPT (`cyipopt`)	no Windows build for Python 3.14; would need conda or an older Python
ParOpt	not on PyPI
TOFLUX	the repository states no licence, so it cannot be reused
MATLAB bridge (run the originals as whole methods)	the MATLAB Engine for R2026a supports Python ≤ 3.13; GNU Octave is the free alternative

References retained from the original scope

Amstutz & Andra (2006): standalone geodesic topological-gradient algorithms remain a candidate; the implemented HJ/TD variant is described in the catalogue. NLopt documentation remains the upstream guide for its wrapped optimisers.